USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: USB3 Gen T Error Source Clarifications
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes :
• Clarification to the list of the USB3 Gen T error causes and expected behaviors.

Benefits as a result of the changes:
Clearer description of the Gen T link layer error handling, with alignment to Section 7.3 of the USB3 Base Specification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
NA

An analysis of the hardware implications:
NA

An analysis of the software implications:
NA

An analysis of the compliance testing implications:
NA

Actual Change

(a). Section 9.4.1.6.5. LTSSM.ERROR

From Text:

A USB3 Gen T Port shall enter the LTSSM.ERROR state if any of the following occur:

- Expired *tPortReadyTimeout* (Link Ready handshake failure).
- Expired PENDING_HP_TIMER or expired CREDIT_HP_TIMER .
- Packet decode error (Protocol Error).
- LGOOD_n with unexpected Header Sequence Number.
- Link Command Word Header Sequence Number Error.
- Reception of OOBM.*LinkError* (only for USB3 Gen T downstream port).

When Link Keep-Alive is enabled, a Gen T downstream port in LTSSM.U0 state may enter LTSSM.ERROR as described in Section 9.4.1.5.1.

To Text:

A USB3 Gen T Port shall enter the LTSSM.ERROR state ~~if any~~ by~~of~~ the ~~following occur~~errors listed in 9.4.1.6.8 which leads to transitioning to the LTSSM.ERROR state.

A USB3 Gen T downstream port in LTSSM.U0 state shall enter LTSSM.ERROR state on reception of OOBM.*LinkError*.~~:~~ When Link Keep-Alive is enabled, a Gen T downstream port in LTSSM.U0 state may enter LTSSM.ERROR as described in Section 9.4.1.5.1.

- ~~Expired *tPortReadyTimeout* (Link Ready handshake failure).~~
- ~~Expired PENDING_HP_TIMER or expired CREDIT_HP_TIMER .~~
- ~~Packet decode error (Protocol Error).~~
- ~~LGOOD_n with unexpected Header Sequence Number.~~
- ~~Link Command Word Header Sequence Number Error.~~
- ~~Reception of OOBM.*LinkError* (only for USB3 Gen T downstream port).~~

(b). Section 9.4.1.6.8. Error Checking

From Text:

An Internal USB3 Gen T Component shall process errors using the error checking mechanism and flow defined in the USB 3.2 Specification. However, there shall be no recovery for errors in Link Command transactions. Upon detecting an error or a timeout, the USB3 Gen T Port shall transition to the LTSSM.ERROR state.

Table 9-11. Error Detection and Handling

Error	Action Taken by the Detecting USB3 Gen T Port	Detecting the Error by the Link Partner
PENDING_HP_TIMER Timeout	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
		DFP: Reception of OOBM.*LinkError*
LGOOD_n with unexpected Header Sequence Number	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
		DFP: Reception of OOBM.*LinkError*
CREDIT_HP_TIMER Timeout	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
		DFP: Reception of OOBM.*LinkError*
PM_ENTRY_TIMEOUT	Transition to LTSSM.Ux state	None
PM_LC_TIMER Timeout	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
		DFP: Reception of OOBM.*LinkError*
Link Command Word Header Sequence Number Error	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
		DFP: Reception of OOBM.*LinkError*

To Text:

An Internal USB3 Gen T Component shall process errors using the error checking mechanism and flow defined in the USB 3.2 Specification. However, there shall be no recovery for errors in Link Command transactions. Upon detecting an error or a timeout, the USB3 Gen T Port shall transition to the LTSSM.ERROR state as mentioned in Table 9-11.

Table 9-11. Error Detection and Handling

Error	Example / Description	Action Taken by the Detecting USB3 Gen T Port	Detecting the Error by the Link Partner
Link Ready handshake failure	Link Ready timeout (tPortReadyTimeout)	Transition to LTSSM.ERROR state	Subsequent Link Ready handshake completion timeout (tPortReadyTimeout)
Port Reset handshake failure	Port Reset timeout (tPortReadyTimeout)	Transition to LTSSM.ERROR state	Subsequent Link Ready handshake completion timeout (tPortReadyTimeout)
Inferred UFP Disconnect (DFP only)	Host did not detect LKAN within the Wait Interval while in U0 state	Optional transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
UFP Error Message (DFP only)	Reception of OOBM.*LinkError*	Transition to LTSSM.ERROR state	None (UFP already in LTSSM.Error State)
PENDING_HP_TIMER Timeout	LGOOD_n not received upon PENDING_HP_TIMER timeout	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout DFP: Reception of OOBM.*LinkError*
Rx Header Sequence Number Error	The Header Sequence Number in the received header packet exceeds the expected Rx Header Sequence Number, or Rx Header Buffer is not available to store the packet	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout DFP: Reception of OOBM.*LinkError*
Type 1/Type 2 Rx Buffer Credit Error	LCRD1_x or LCRD2_y was received out of order	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout DFP: Reception of OOBM.*LinkError*
LGOOD_n with unexpected Header Sequence Number	The Header Sequence Number in the received LGOOD_n does not match any of the outstanding ACK Tx Header Sequence Numbers	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout DFP: Reception of OOBM.*LinkError*
CREDIT_HP_TIMER Timeout	LCRD1_x or LCRD2_y not received upon CREDIT_HP_TIMER timeout.	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout DFP: Reception of OOBM.*LinkError*

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Error	Example ~~/ Description~~	Action Taken by the Detecting USB3 Gen T Port	Detecting the Error by the Link Partner
PM_ENTRY_TIMEOUT	Timeout after sending LAU without receiving LPMA	Transition to LTSSM.Ux state	None
PM_LC_TIMER Timeout	Timeout after sending LGO_Ux without receiving neither LAU nor LXU	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
			DFP: Reception of OOBM.*LinkError*
Invalid Link Command Word ~~Header Sequence Number Error~~	Identifying an invalid link command word within a Gen T Link Command Tunneled Packet	~~Transition to LTSSM.ERROR state~~Ignored	~~UFP:~~ Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
			Future Link Command Word may not be supported~~DFP: Reception of OOBM.*LinkError*~~
Invalid OOBM	Identifying an invalid OOBM Tunneled Packet Payload if CRC check is correct.	~~Ignored?~~Ignored	~~TBD~~Transitioning to LTSSM.Error state on timeout, or no impact
Data Packet Framing Error	Packet is missing END/EDB symbols	Transition to LTSSM.ERROR state	UFP: Subsequent transaction results in PENDING_HP_TIMER or CREDIT_HP_TIMER or PM_LC_TIMER Timeout
			DFP: Reception of OOBM.*LinkError*